FINANCE STOCK DATA ANALYSIS REPORT

AUTHOR : Ayisha Mariyam

PROJECT OUTLINE :

This project focuses on analyzing stock market data to understand price behavior, trading activity, and relationships between key financial indicators. The analysis follows a structured data analytics workflow to extract meaningful insights from the dataset.

PROJECT SUMMARY :

Financial markets produce large volumes of data that can be analyzed to identify patterns, trends, and risks associated with stock trading. This report presents an exploratory analysis of stock market data to understand the behavior of stock prices and trading activity. The dataset includes key financial indicators such as opening price, closing price, highest price, lowest price, and trading volume. Using Python-based data analysis tools, the dataset was cleaned, explored, and visualized to identify relationships between these variables. The analysis reveals that stock prices fluctuate significantly over time, reflecting normal market volatility. Strong relationships were observed between opening and closing prices as well as between high and low prices. Trading volume patterns also indicate variations in investor activity across trading periods. These insights demonstrate how financial data analysis can support investment decisions, market monitoring, and risk management strategies.

METHODOLOGY :

Before conducting the analysis, the dataset was prepared to ensure data accuracy and reliability. The following preprocessing steps were performed:

- The dataset was loaded using the **Pandas** library in Python.
- The dataset structure was examined using .info() and .describe() functions.
- Missing values were checked using .isnull().sum().
- Rows containing missing values were removed to maintain data consistency.
- Duplicate records were identified and removed using .drop_duplicates().
- Numerical columns such as price and volume were verified to ensure they were in the correct format for analysis.

These steps ensured that the dataset was clean and ready for reliable analysis.

TOOLS USED :

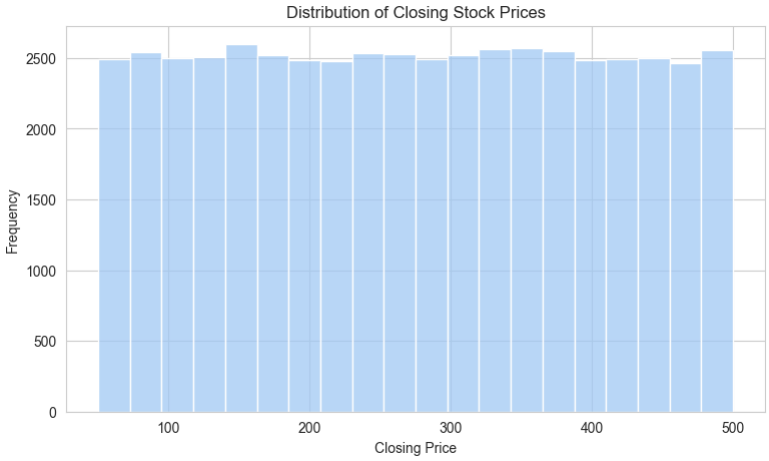
Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook were used for performing data cleaning, analysis, and visualization.

ANALYSIS FINDINGS (KEY ANALYSIS AREAS) :

Several visualizations were created to understand the behavior of stock prices and trading activity.

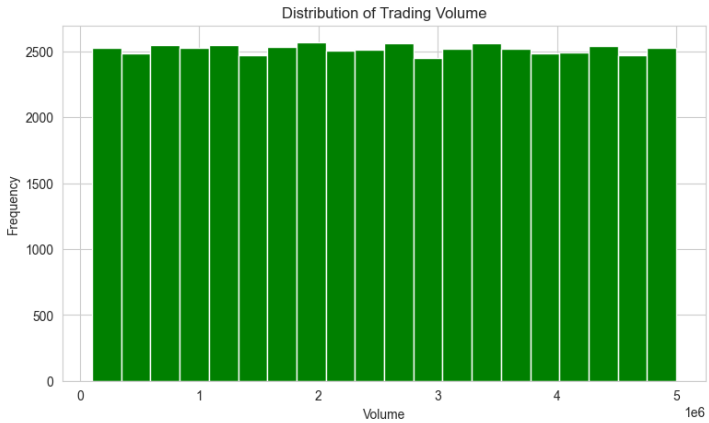
- **Distribution of Closing Stock Prices :**

  A histogram of closing stock prices was generated to analyze how stock prices are distributed across the dataset. The visualization shows that most stock prices fall within a moderate range, while fewer instances appear at very high or very low values.



- **Trading Volume Distribution :**

  The trading volume histogram shows the frequency of stock trades. The results indicate that trading volume varies significantly, suggesting fluctuations in market participation and investor interest.



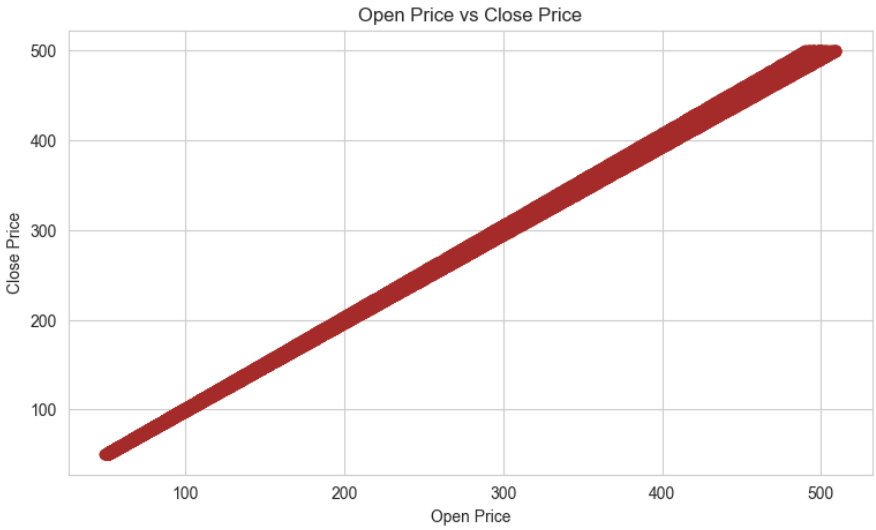
- ## Stock Price Trend:

  A line chart was used to visualize the closing price trend over time. The trend shows fluctuations that represent typical market movements influenced by supply, demand, and external economic factors.

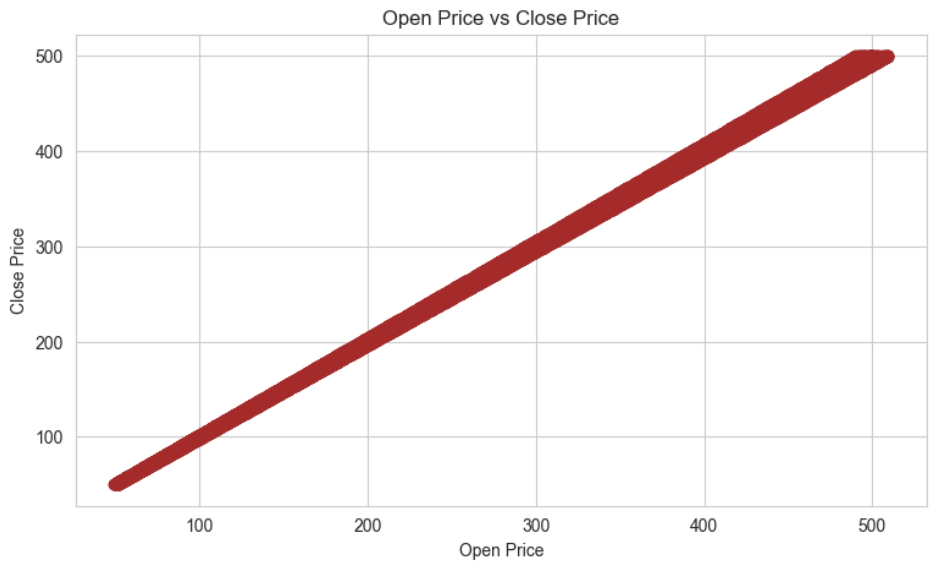
- ## Price Relationship Analysis :

  Scatter plots were used to examine relationships between different price indicators.

  - **Open vs Close Prices:** These variables show a strong relationship since closing prices are typically influenced by opening prices and market activity throughout the trading day.



- **High vs Low Prices:** These values represent the daily trading range and are strongly correlated.



STATISTICAL INSIGHTS :

Statistical analysis provides a deeper understanding of stock price patterns and variability.

Key statistical observations include:

- The **average closing price** indicates the typical stock price level within the dataset.
- **Standard deviation values** show the degree of price variability, reflecting stock market volatility.
- **Minimum and maximum values** highlight the range of stock price movements.
- **Correlation analysis** indicates strong relationships between stock price indicators such as open, close, high, and low values.
- **Daily return calculations** show fluctuations in stock performance across trading periods.

These statistical insights help evaluate stock performance and market stability.

KEY FINDINGS :

- Stock prices show noticeable fluctuations over time.
- A strong positive relationship exists between opening and closing prices.
- High and low prices are strongly correlated.
- Trading volume varies significantly across trading periods.
- Daily return analysis reveals moderate volatility.
- Price-related variables (Open, High, Low, Close) demonstrate strong correlations
- Periods of increased trading volume often coincide with higher price movements

BUSINESS RECOMMENDATIONS :

- Monitor Market Volatility.
- Use Correlation Insights for Strategy Development.
- Track Trading Volume Trends.
- Apply Data-Driven Investment Decisions.
- Implement Risk Management Practices.

CONCLUSION :

This report presented an exploratory analysis of a stock market dataset to understand stock price behavior and trading patterns. The analysis involved data cleaning, exploratory visualization, correlation analysis, and statistical evaluation. The results show that stock prices demonstrate natural market volatility and strong relationships between key price indicators. Trading volume patterns also highlight variations in market activity over time. Financial data analysis plays a critical role in supporting investment strategies, risk assessment, and market monitoring. By leveraging data analytics tools, investors and analysts can gain valuable insights that contribute to informed decision-making. Future work may include advanced financial modeling techniques such as time-series forecasting, machine learning-based stock prediction, and portfolio optimization.